

Transit Signal Priority (TSP) User Guide For Advanced System Controller ASC/3

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1. INTRODUCTION

1.1 Background

Urban congestion has led many jurisdictions to coordinate traffic signals along heavily traveled arterials, in order to decrease delays and operating costs for vehicles. However, this type of coordination has not solved numerous problems encountered by transit vehicles in congested areas, such as delays, schedule inconsistency, increased operating costs and trip times. Buses do not operate on the same cycle or pattern as normal passenger vehicles due to their frequent stops, so they require a different solution.

Traditional solutions to expediting transit vehicle traffic have been based on preemption, which frequently involves skipping intersection phases and sacrificing system coordination for lengthy periods. The uncoordinated delay and congestion resulting from the use of preemption can last up to seven coordination cycles (one preemption cycle plus six cycles to return to coordination), approximately seven minutes (assuming a 60-second cycle) for each transit vehicle preemption.

1.2 The Transit Signal Priority (TSP) Solution

Econolite Control Products, Inc. (Econolite) has worked for years with agencies and consultants to design controller features and options that address transit problems. The most complete solution available for transit optimization is based on the new Econolite Transit Signal Priority (TSP) module for the ASC/3 Series controller. TSP yields considerable advantages over previously available methods; during TSP operation, there is no interruption of coordination and no omission of phases.

Even though an ASC/3 controller has the software and screens to operate TSP, you must enable TSP in order to activate the TSP function. For TSP operation, you need to insert a Special Feature blue Datakey® and keep it in its slot. To enable TSP, use the procedure given in **Para 3.1**. Once it is enabled and configured, TSP operation goes into effect automatically when a transit vehicle is detected. It maximizes throughput for both transit and private vehicles, and provides the following key benefits:

- Improved transit vehicle schedule consistency
- Decreased transit vehicle route travel times
- Reduced negative impact of transit priority on private vehicles
- Maximized personal throughput for transit and private vehicles.

1.3 Purpose of this Guide

This guide describes the TSP functionality, in particular how to input data using the ASC/3 controller front panel menus. This same programming can also be accomplished using the comparable menu screens in the Aries system software.

1.4 Conventions

Table 1-1 lists the text conventions in this user guide.

Table 1-1. Text Conventions

Convention	Description	Examples
References to tables and paragraphs.	Bold text. This is a hyperlink field.	Paragraph 1.3
Keys to press in a procedure	Square brackets enclose the name of the key	Press [7] .
Titles of screens and fields in screens	Uppercase bold	TSP SPLIT PTN DELAY TIME
Navigation names of screens	Bold “ MM ” (MAIN MENU) followed by the keys to press to reach the screen from the Main Menu.	MM-1-6-1

1.5 Screen Navigation

The configuration instructions in this user guide are for the ASC/3 Controller front panel keys and screen. However, you can also use these same instructions to configure the ASC/3 for TSP with the PC-based Database Editor; the hierarchy of screens is the same, and the screens have the same names. For example, to access **MM-4-3**, the main TSP screen, the procedure with the controller front panel is:

1. Press **[MAIN MENU]** to go to the **MAIN MENU (MM)**.
2. Press **[4]** to go to the **PREEMPTOR/TSP/SCP SUBMENU**.
3. Press **[3]** to go to the **TSP/SCP PLAN 1-6** display.
4. **MM-4-3, TSP/SCP PLAN**, comes into view (for details, refer to **Paragraph 3.4.4**).

With the Database Editor on a PC, you also access **MM-4-3** from the **MAIN MENU** when you click through the same hierarchy of screens. In some cases, more than one screen is in view on the PC monitor, but they are labeled the same as the controller front panel screens.

1.6 Related Documentation

Please refer to the following manuals and website URLs for additional information.

- *Advanced System Controller ASC/3 Programming Manual* (Part No. 100-0903-001). Provides architectural, hardware and comprehensive programming information for the Econolite ASC/3 controller product line.
- NTCIP Data Manager Guide
- <http://www.econolite.com>. Provides data sheets, white papers, and Technical Support contact information for Econolite Control Products, Inc.

Introduction

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2. TSP THEORY OF OPERATION

2.1 The TSP Cycle

TSP operation takes place in several steps. The most significant step is the TSP Active Sequence when TSP adjusted signaling is taking place. The steps in the TSP operational cycle are:

1. **Check-in** – Input to the controller from priority detection that senses the arrival of the transit vehicle and initiates the TSP functionality in the controller.
2. **TSP Delay** – Controller timing function that can be set to delay the beginning of TSP operation until a set interval after check-in. Each illustration in **Figure 2-1** thru **Figure 2-4** shows a delay interval. Set the **DELAY TIME** in **MM-4-3**.
3. **Begin TSP Active Sequence** – Controller begins using the TSP timing data for improved transit operation, as configured using the controller front panel menus.
4. **Check-out** – Input to the controller from priority detection that determines the departure of the transit vehicle (optional, TSP cycle can end without a check-out).
5. **End TSP Sequence** - The end of the TSP routine is based on check-out, timeout or lock released. Lock (**DET LOCK**) and the timeout interval (**MAX PRESENCE**) are configured in **MM-4-3**.

There are a number of ways to sense priority detection. The controller currently supports Hardware Input priority detection that can come in as **SOLID** or **PULSING** signals. It also supports Metro Rapid Protocol over a serial port.

2.1.1 Metro Rapid Protocol

Metro Rapid Protocol is supported on Port 2 of the ASC/3 controller for TSP operation, as priority messages.

How to enable Metro Rapid:

1. Go to **MM-1-5-2**.
2. Move the cursor right to the **PROTOCOL** field.
3. Toggle (press **[0]** again and again) until **METRO RAPID** comes into view.
4. Priority messages will be treated as **SOLID** inputs as described in **Paragraph 2.1.2** below.

2.1.2 Single Pulse (SOLID) Check-In / Check-Out TSP Cycle

When the transit vehicle is detected, a single-pulse signal is applied to the TSP Check-In input on the ASC/3 controller. This input causes a TSP Check-In, followed by the delay timer, followed by the start of the TSP Sequence. When the transit vehicle clears the intersection a single-pulse is applied to the TSP Check-Out pin on the controller. This causes a TSP Check-Out, followed by the end of TSP Sequence. Use **MM-4-2** to map TSP inputs; refer to **Paragraph 3.3.1**. (Also, the Check-In and Check-Out functions can be individually assigned to any unused pin using the ASC/3 Configurator. Contact your Econolite representative for instructions and assistance.)

Figure 2-1 shows a typical TSP cycle with Single Pulse (SOLID) check-in and check-out.

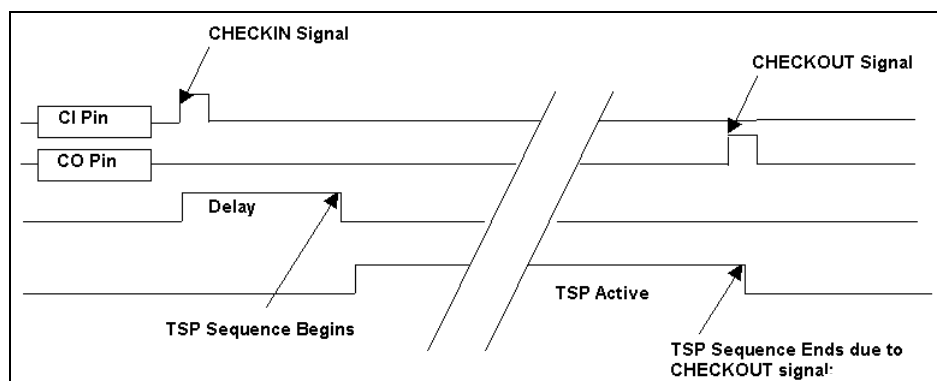


Figure 2-1. TSP Cycle Ending with Single Pulse Check-Out

TSP operation that begins with a single-pulse check-in typically ends due to a single pulse check-out, as shown in **Figure 2-1**. However, the TSP cycle can also end due to a timeout condition if a check-out signal is not received within the cycle time set for normal operation, as shown in **Figure 2-2**.

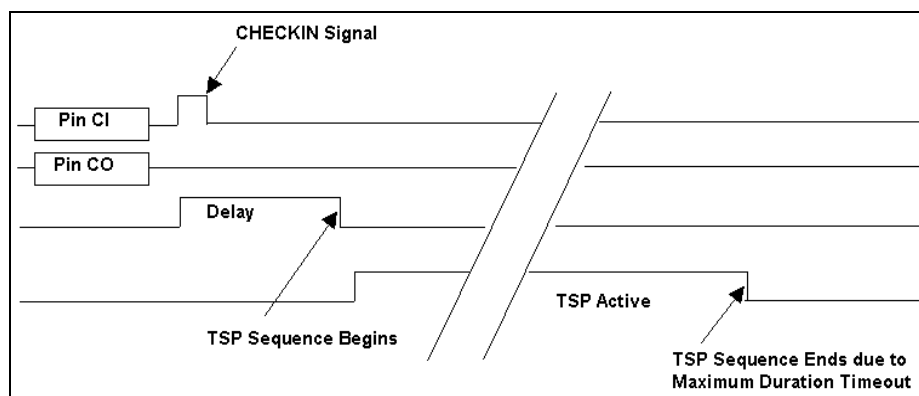


Figure 2-2. TSP Cycle Ends in Timeout (no Check-Out)

2.1.3 Continuous Pulses (PULSING) TSP Cycle

Check-in and check-out can also be based on a call from an optical preemption system such as Opticom™. These are often used for transit or emergency vehicle detection at an intersection. A check-in occurs when an optical sensor picks up a request from a transit vehicle, and applies a continuous 6.25 Hz signal (low-priority call) to the controller.

Once the TSP Check-In is recognized, the delay timer begins timing, followed by the beginning of the TSP Sequence. Once the 6.25 Hz signal ends, a check-out takes place. The TSP Sequence ends under one of three conditions:

- 1) At the Check-Out (lifting of the 6.25 Hz input);
- 2) The start of the TSP phase(s) (refer to **Paragraph 3.4.4**, prompt **DET LOCK**, and “Detector Lock — ECPI” in **Table 2-1** with respect to locking); or
- 3) At the calculated maximum timer (internally calculated based on the normal pattern cycle length).

In the *first condition*, upon lifting of the 6.25HZ input, the TSP sequence is immediately terminated when TSP is set to non-lock. Refer to **Figure 2-3** and **Paragraph 3.4.4**, prompt **DET LOCK**.

In the *second condition*, TSP cycle ending will take place upon serving the TSP phase(s), and will override a Check-Out (lifting of the 6.25 input) until the TSP phase(s) is served. This function assures the early green return to the TSP phase(s). If an early check-out takes place before the Lock is released, then the TSP cycle will continue on until the lock is released, as shown in **Figure 2-4**.

Refer to “Detector Lock — ECPI” in **Table 2-1**. If no detector lock is configured, and the Check-Out is received before the Delay is complete, TSP demand will end.

In the *third condition*, the TSP cycle may end due to a timeout condition if the maximum duration is reached before a check-out takes place. **Figure 2-3** shows the TSP cycle with steady pulsing check-in and check-out.

TSP Theory of Operation

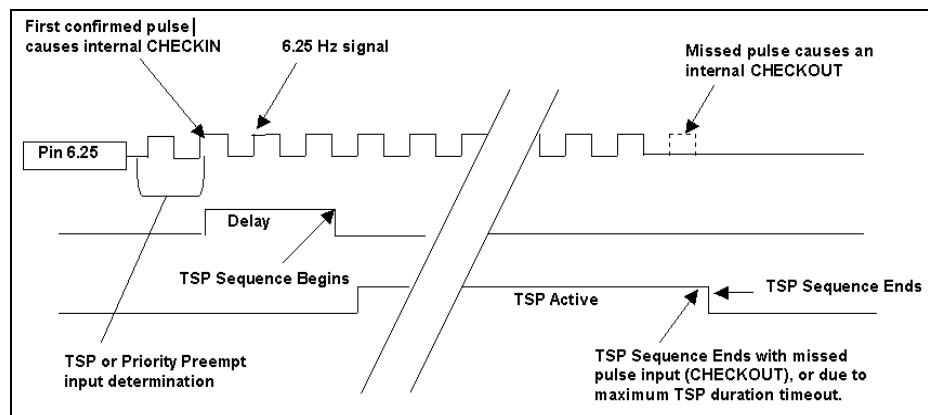


Figure 2-3. TSP Cycle with Continuous 6.25 Hz Pulse

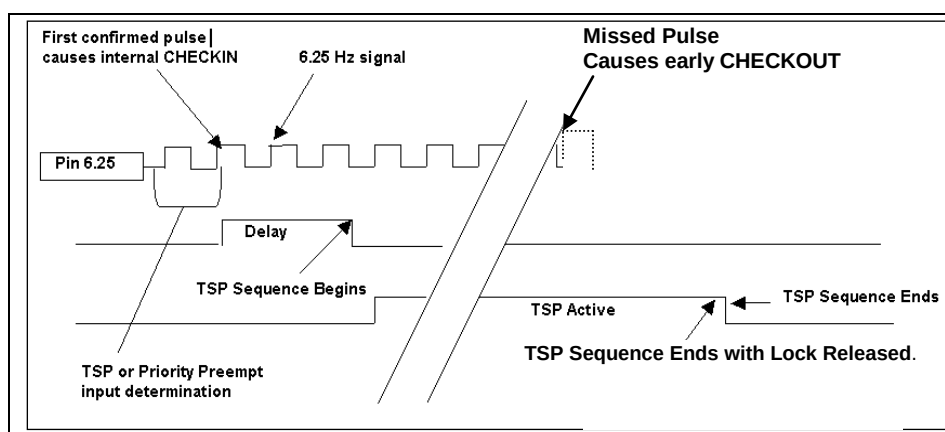


Figure 2-4. TSP Cycle Ending with Lock Released

2.2 TSP Active Sequence Patterns

An ASC/3-series controller supports up to 120 coordination patterns. When using the TSP module, alternate coordination patterns (TSP patterns) are called/linked during a TSP sequence to optimize service for the approaching transit vehicle. Each normal coordination pattern can be linked to a unique or shared TSP pattern, depending on how the controller's coordination patterns are being used.

Each TSP Pattern provides “split” times for all TSP phase(s). Non-TSP phase(s) use the TSP Pattern entry, whereas the ASC/3 controller calculates TSP phase(s) times. Use **MM-3-2** and **MM-4-4** to configure TSP Patterns (refer to **Chapter 3** for detailed instructions)

When a transit vehicle is detected approaching the intersection and a TSP Check-In occurs, a particular TSP pattern is selected to optimize bus transit. The selection occurs differently depending on whether the intersection is operating in Coordinated or Free mode.

- *Intersection in Coordination Mode:* The current coordinated pattern transfers control to the linked TSP pattern once a TSP check-in takes place. When the TSP active sequence ends, the initial coordination pattern (pattern running prior to TSP commencement) regains control. Normally, the controller uses the coordination pattern in effect during transit hours and modifies the splits to a TSP pattern with values from **Screen 4-4, TSP/SCP SPLIT PATTERN**.
- *Intersection in Free Mode:* A TSP check-in will automatically switch the current phase MAX green time to its TSP free pattern split data field.
 - With **MM-4-3**, use **FREE DEFAULT PTN** to select any pattern (1 thru 120) for the Free Mode default pattern.

Only one transit vehicle can modify timing during any given cycle. Transit Priority operation is on a first-come first-serve basis. Service of another transit vehicle demand may be prohibited during the cycles that follow a modified cycle. Use **MM-4-3, RESERVICE CYCS** to enter the number of cycles to not allow another TSP cycle after a TSP cycle.

TSP operation takes place in conjunction with the normal controller functions. TSP is not a preempt function.

As such, TSP operation is overridden and inhibited by:

- Railroad or emergency vehicle preemption.
- Coordination offset seeking.
- Time Base Special Function to inhibit any number of the 6 TSP inputs (**MM-4-3, ACTION SF INHIBIT** and **MM-5-2, SPC FCT**).
- TSP phase(s) omitted or skipped.
- One TSP phase leads a coordinated phase while the other TSP phase lags.
- A TSP sequence has modified the phase splits during coordination and TSP operation is inhibited during the rest of that cycle and the reservice cycles.

2.2.1 TSP Active Sequence in Coordinated Mode

A TSP active sequence that is initiated when the controller is coordinated does not interrupt coordination or omit phases. Minimal time is required to transition into and out of TSP operation because of the use of “check-in”-“check-out” operation. The TSP sequence is not a preemption of the intersection; it is a modification of the phase splits to benefit the transit movements.

Figure 2-5 shows a pie-chart representation of a Coordinated Cycle that includes a TSP sequence with no timing changes of the basic coordinated pattern. The transit vehicle, requiring phase 2&6 initiates the check-in and check-out process during the guaranteed green period of coordinated phase(s) 2/6.

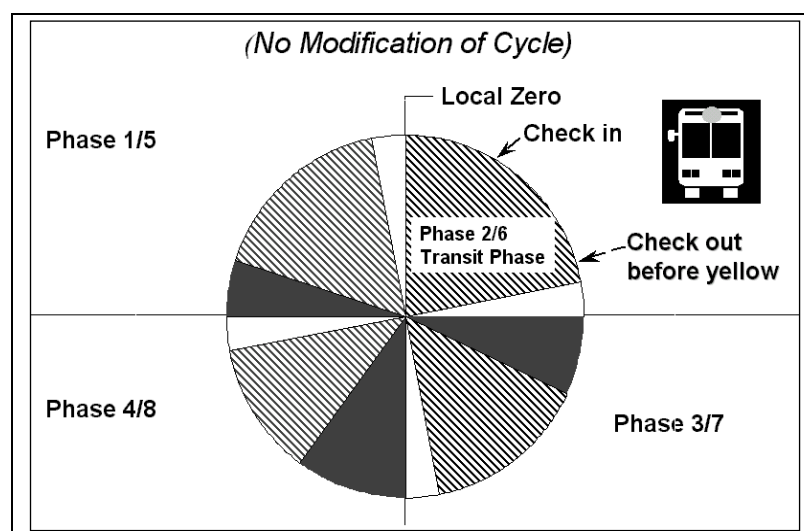


Figure 2-5. Coordinated Cycle Unaffected by TSP Active Sequence

Legend:

Black Slices = Minimum green interval(s) configured by the TSP pattern phase splits.

Striped Slices = Coordinated phases (e.g. 2/6) green interval and extendable green time for the non-coordinated phases.

White Slices = Clearance yellow and red intervals.

Figure 2-6 shows the maximum phase length that can be assigned to phases 2/6 during the TSP active sequence, before the TSP sequence ends in a timeout, not check-out.

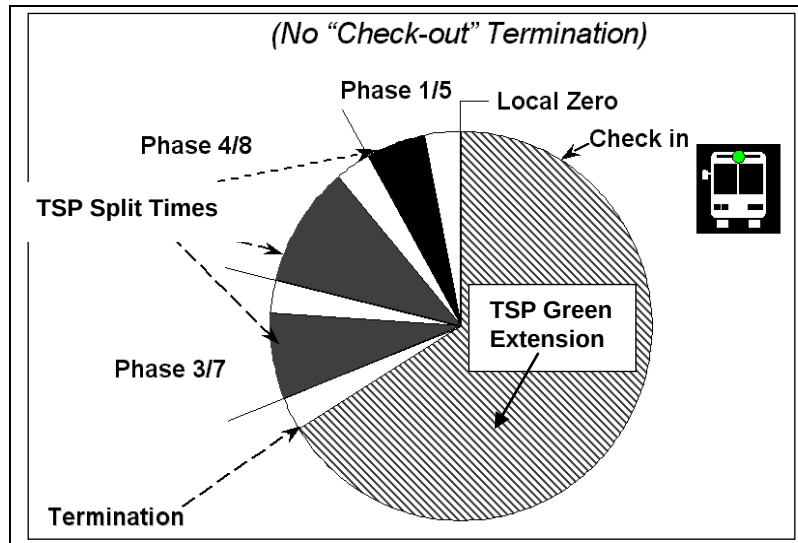


Figure 2-6. Coordinated Cycle with Maximum Length TSP Transit Phase

Legend:

Black Slices = TSP Split Time configured by the TSP pattern phase splits.

Striped Slices = TSP Green interval and Extensions.

White Slices = Clearance yellow and red intervals.

The phase splits configured in the TSP Pattern Data indicate the minimum splits allotted to each non-transit phase during TSP operation. The timeout termination (maximum TSP phase Extension) shown in **Figure 2-6** ensures that a minimum interval is allowed for each phase within the coordinated cycle time.

Figure 2-7 shows a coordinated cycle that encounters a transit vehicle requiring service on phase(s) 4/8, but arriving for check-in during phase(s) 2/6. Phase(s) 2/6 and phase(s) 3/7 are decreased to the minimum interval allowed by the TSP Pattern, and phase(s) 4/8 is extended until the check-out occurs.

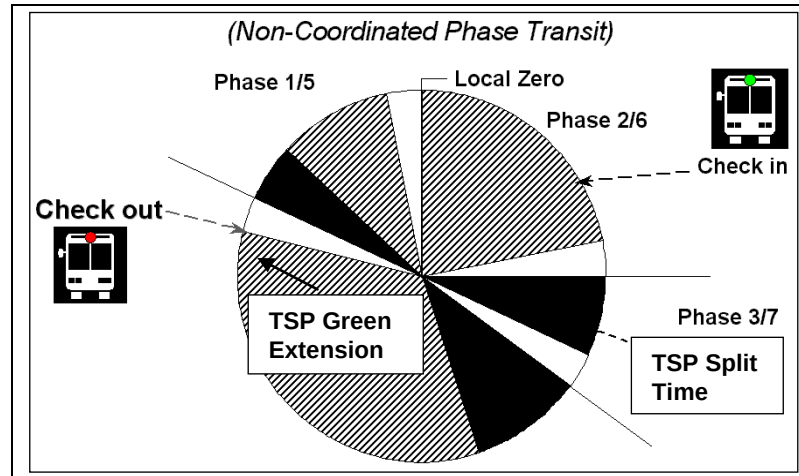


Figure 2-7. Check-In Occurs Before the Transit Phase

Legend:

Black Slices = TSP Split Time configured by the TSP pattern phase splits.

Striped Slices = TSP Green interval and Extensions.

White Slices = Clearance yellow and red intervals.

2.2.2 TSP Active Sequence in Free Mode

When the intersection is operating in a Free Mode, there is no set cycle time/length. In response to a TSP check-in (of the 6 possible TSP Inputs), the controller replaces the Non-TSP phase maximum green times with the times (in seconds) that are entered in the split table of the Free Pattern for that TSP Input.

- With **MM-4-3**, enter the number of the **FREE DEFAULT PTN**.

The TSP phase maximum green (in seconds) is the cycle length (30-999, MM-3-2) entered into the Free Pattern minus the sum of the Non-TSP phase minimum green times (in seconds, MM-4-4) and clearance times (in seconds, MM-2-1) entered in the Free Pattern and the clearance time of TSP phase. TSP phase minimum green time is calculated by subtracting the phase maximum reduction time in TSP split pattern from the TSP phase split in Free Pattern.

EXAMPLE: Standard Quad with phase 1-4 in ring 1 and 5-8 in ring 2

Cycle length of free pattern (MM-3-2) = 100s

All phases have the same timings:

Phase Clearance time (MM-2-1) = 5s

Phase split in Free Pattern (MM-3-2) = 20s

Max Reduction time (MM-4-4) = 10s

TSP phase minimum green = Phase split in Free Pattern – Max Reduction Time
 $= 20s - 10s = 10s$

TSP phase maximum green = Cycle length – non-TSP phase min green and clearance times
 – TSP phase clearance time
 $= 100 - (10 + 5) * 3 - 5 = 50s$

Phases are not skipped during a TSP operation unless a phase(s) omit is active during TSP operation.

Figure 2-8 shows TSP operation during the Free mode, with a check-in occurring during phase(s) 2/6, and the transit vehicle service and checkout occurring during phase(s) 4/8. Phase(s) 3/7 is decreased to a minimum TSP green interval, but no phases are skipped.

Note: Phase 2&6 can be reduced to serve 4&8 earlier if the Free Pattern values entered are less than the normal Max green value.

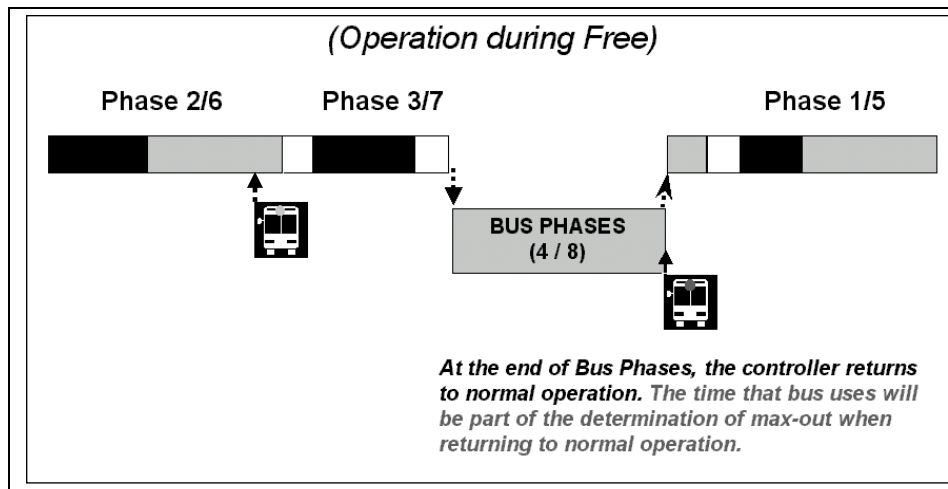


Figure 2-8. TSP Operation in Free Mode

Legend:

Black Bars = TSP phase Green Maximum times as configured by the TSP Free Pattern phase splits.

Light Gray Bars = Phase 2/6 green extension intervals.

= Phase 4/8 is able to extend to its maximum green after TSP

Checkout.

= Phase 1/5 normal green extension interval.

White Bars = Clearance yellow and red intervals.

2.3 TSP Operational & Timing Parameters

Table 2-1 provide detailed information on the TSP Configuration Parameters that are included as part of the TSP module functionality. Refer to **Chapter 3** for step-by-step instructions to configure the ASC/3 Controller data entry screens for TSP. The tables reference the screens to use.

Table 2-1. Configuration Parameters

Configuration Parameters	Screen Navigate	Screen Prompt	Range	Default
TSP Plan Enable Option This is a per TSP plan parameter: Toggle to select: NO: Disable TSP/SCP operation called from this input. YES: Phasing as programmed in (MM-1-1-1) and (MM-1-1-2). NOTE For information about Diamond Intersections, refer to the <i>ASC/3 Programming Manual</i> (100-0903-001), Appendix Q/Chapter 29, <i>Diamond Intersection Application Notes</i>.	MM-4-3	TSP/SCP ENA	NO, YES	NO
Detector Signal – ECPI This is a per TSP plan parameter that selects the Check-In detector signal to be solid or pulsing. NOTE: Priority messages over serial or over NTCIP will be treated as Solid Input. P – TSP pulsing pins are 6.25HZ pulsing signals that cause a Check-In when the pulsing starts and a check-out when the pulsing stops. S– TSP Check-In pins are single-pulse signals. TSP Check-Out pins stop and prohibit TSP operation.	MM-4-3	SIGNAL TYPE	S, P	S

Configuration Parameters	Screen Navigate	Screen Prompt	Range	Default
Detector Lock – ECPI This is a per TSP plan parameter that allows a momentary input to be locked if the input is set to PULSING. X – Momentary pulsing input will be locked. TSP will be released under the following conditions: 1. TSP phases are active: when delay terminates, then TSP will release at the end of TSP phase extension period, or when a check-out received. 2. TSP phases are not active: when delay terminates, then TSP is released at the beginning of the TSP phase(s). . - Momentary pulsing input will not be locked. When inputs are set to SOLID, the Check-In is locked until a Check-Out is received or the TSP operation forces a Check-Out.	MM-4-3	DET LOCK	“.” or X	“.”
Delay Time – ECPI This is a per TSP plan parameter that specifies number of seconds between the TSP Check-In and the start of the TSP movement.	MM-4-3	DELAY TIME	0-255	0
MAX Time – ECPI This is a per TSP plan parameter that specifies the max time in seconds a TSP call can be held or that a TSP movement can be active before it times out.	MM-4-3	MAX PRESENCE	0-999	0
Preempt Enables Reservice – ECPI This is a per TSP plan parameter: “.” – TSP reservice timer will not start when preemption is terminated. X – TSP reservice timer will start when preemption is terminated.	MM-4-3	PMT ENA RESERVICE	“.” or X	“.”

Configuration Parameters	Screen Navigate	Screen Prompt	Range	Default
No Delay When in TSP Phases – ECPI This is a per TSP plan parameter: X - No delay time when a TSP phase is timing in green. “.” – Call will not be serviced when delay time is timing.	MM-4-3	NO DELAY IN TSP/SCP	“.” or X	“.”
Action Plan SF Inhibit Output - ECPI This is a per TSP plan parameter in the range 1-8 that specifies which timebase special function output (programmed in MM-5-2 – Action Plan) is to inhibit the associated TSP input.	MM-4-3	ACT SF INHIBIT	0-8	0
Reservice Cycles – ECPI This is a per TSP plan parameter that specifies the number of cycles that all TSP inputs will be inhibited following a TSP movement or when any phase max or split has been altered. Coordinated mode: The cycle length is determined from the coordinated pattern in effect. Free mode: The cycle length is determined from the selected Free pattern. The reservice operation will inhibit all TSP operations until it is complete.	MM-4-3	RESERVICE CYCS	0-255	0

TSP Theory of Operation

Configuration Parameters	Screen Navigate	Screen Prompt	Range	Default
Select TSP or SCP – ECPI or NTCIP TSP – ECPI TSP operation is enabled. TSP coordinated split values are calculated by modifying the split values of MM-3-3 with values entered in MM-4-4 in the TSP/SCP MAX RDTN fields. For free operation, you specify maximum green time. SCP – NTCIP SCP operation is enabled. During SCP operation the max and min phase split or MAX green times are determined as follows: Max phase split = phase split in SCP split pattern + phase MAX extension Min phase split = phase split in SCP split pattern – phase MAX reduction	MM-4-3	TSP OR SCP	TSP or SCP	TSP
Default Pattern to use when in Free Mode. NOTE: This is supported only if TSP FREE PTN = 0 in MM-4-3.	MM-4-3	FREE DEFAULT PTN	1-120	120
BUS HEADING to TSP inputs 1-4 mapping for MTA priority request (Only applicable with Metro Rapid Protocol that you set in MM-1-5-2)	MM-4-3	BUS HEADING	NB SB EB WB	TSP 1 = NB TSP 2 = SB TSP 3 = EB TSP 4 = WB
Bus Headway Allowance for MTA priority request. (Only applicable with Metro Rapid Protocol that you set in MM-1-5-2) This specifies the percentage of headway allowance that the controller will response to the MTA priority request	MM-4-3	HEADWAY ALLOWANCE	0-255	0%

Configuration Parameters	Screen Navigate	Screen Prompt	Range	Default
TSP Phase Assignment. For each TSP/SCP plan, each phase can be assigned to one of the options below: . = No programming T = TSP/SCP Phase. referred to as Priority Strategy Service (TSP) Phases – NTCIP 1211 4.2.5.1.2 V = Phase Omit. This is a per TSP plan parameter. The Coordinator shall ignore any omit applied to the coordinated phase(s). This phase omit is logically ORed with any other phase omits that may be active. Refer to NTCIP 1211 4.2.5.1.3 P = Ped Omit. This is a per TSP plan parameter. This ped omit is logically ORed with any other ped omits that may be active. These ped omits do not apply to non-actuated phases. Refer to NTCIP 1211 4.2.5.1.4	MM-4-3	TSP/SCP PHASE rows	“.” = No programming T = TSP/SCP Phase V = Phase Omit P = Ped Omit	“.”
SCP Phase Max Extension – NTCIP 1211 4.2.6.1.2 This is a per-phase and per-split pattern parameter that specifies the maximum number of seconds that the split of a phase can be extended during SCP. The effective value may be modified based on time gained from one cycle of reductions to the respective non-priority phases. NOTE: This is supported only if “TSP OR SCP” option is set to “SCP”	MM-4-4	MAX EXTN	0-255	255

TSP Theory of Operation

Configuration Parameters	Screen Navigate	Screen Prompt	Range	Default
<i>For TSP operation:</i> This is a per-phase and per-split pattern parameter that specifies the minimum split times by subtracting this TSP/SCP MAX RDTN value from the split values in MM-3-3. <i>For SCP operation:</i> Phase Max Reduction – NTCIP 1211 4.2.6.1.1 This is a per-phase and per-split pattern parameter that specifies the maximum number of seconds that the split of a phase can be reduced during SCP. This effective value may be reduced based on the operating phase minimum service requirements.	MM-4-4	MAX RDTN	0-255	0

3. TSP ENABLE AND CONFIGURATION

3.1 Enable the TSP Special Feature

Even though an ASC/3 controller has the software and screens to operate TSP, you must enable TSP in order to activate the TSP function. To enable TSP, use the procedure below.

3.1.1 Procedure to Enable the TSP Special Feature

```

COPY UTILITY
      FROM      >      TO
PHASE TIMING.... . > PHASE TIMING.... .
TIMING PLAN..... . > TIMING PLAN..... .
PH DET OPT PLAN. . > PH DET OPT PLAN. .
DETECTOR PLAN... . > DETECTOR PLAN... .
COORD PATTERN... . > COORD PATTERN... .
SPLIT PATTERN... . > SPLIT PATTERN... .
SEQUENCE..... . > SEQUENCE..... .
DATA KEY..... . > CONTROLLER DATA. .
CONTROLLER DATA. . > DATA KEY..... .
DEFAULT DATABASE . > CONTROLLER DATA. .
CONTROLLER DATA. . > DEFAULT DATABASE. .
MMU PROGRAM..... . > CONTROLLER..... .
SPECIAL FEATURES . > CONTROLLER..... .
      TOGGLE TO SELECT ONE FROM AND TO
      THEN PRESS ENTER
  
```

MM-8-1

Prompt	Step	Procedure
	1	Locate the Datakey® (Part No. 100-1095-501) you received with your TSP order. This contains the TSP Special Feature enable.
	2	Power up the ASC/3.
	3	Insert the Datakey® into the DATAKEY slot in the upper left corner of the front panel of the ASC/3.
	4	Go to MM-8-1, COPY UTILITY and scroll down to the last prompt, SPECIAL FEATURES.
SPECIAL FEATURES	5	Press [0] and toggle to “X” to select.
CONTROLLER (next to SPECIAL FEATURES)	6	Press [0] and toggle to “X” to select.

Prompt	Step	Procedure
	7	Press [ENTER] to enable the programmed special features.
	8	Processing may take a few seconds to complete. The message below shows on the screen: PROCESSING DATAKEY FOR SPECIAL FEATURES
	9	When processing is completed, one of the two messages below comes into view: PROCESSING FOR SPECIAL FEATURES DONE PRESS ANY KEY TO CONTINUE OR NO SPECIAL FEATURES FOUND PRESS ANY KEY TO CONTINUE
	10	To keep the TSP special feature enabled, keep the Datakey[®] inserted in its slot. If you remove the Datakey[®], the ASC/3 will disable the TSP special feature at midnight. NOTE: After you use a Datakey[®] to copy the controller database, it is OK to remove the Datakey[®]. You do NOT need to keep this Datakey[®] in its slot to keep the database settings.

3.2 City and Intersection Code

If it is necessary for your application, you can associate an ASC/3 controller with City and Intersection codes. How to enter City and Intersection Codes in your controller

1. Go to **MM-8-5**.
2. Move the cursor to the **CITY** field and enter the code for the city.
3. Move the cursor to the **INTERSECTION** field and enter the code for the intersection

3.3 I/O Mapping for TSP Operation

To map TSP signals, use the ASC/3 Configurator. If you use Check-In /Check-Out TSP/SCP inputs (**SOLID**) instead of pulsing TSP/SCP inputs (**PULSING**), the field service person that completes the installation can map the Check-In and Check-Out inputs to any input in the controller. There are a possible total of 6 pairs of TSP/SCP I/O functions—6 Check-Ins and 6 Check-Outs. Refer to the *ASC/3 Configurator User Guide for Advanced System Controllers* (for ASC/3 and ASC/3-2070)

(Part No. 100-0903-004).

If you use Preemptor signals, you can also use **MM-4-2** to map TSP signals with SOLID or PULSING; refer to **Paragraph 3.4.1**.

3.3.1 TSP Configuration Instructions

Once the TSP function is correctly enabled, configure the controller for the best operation of the TSP feature for particular intersections and times of day. The remaining paragraphs of this chapter have instructions to configure an ASC/3 for TSP operation.

3.4 Screens to Configure TSP

The TSP software uses some new screens, some modified screens, and some existing screens. To configure the ASC/3 Controller for TSP/SCP operation, use the screens shown in **Table 3-1** with the step-by-step instructions in the referenced paragraphs of this chapter.

Table 3-1. Possible TSP Screens

Screen	Purpose	Para. No.
MM-1-6-1	Enable TSP/SCP Event Logging	3.9
MM-1-8-2	Program Logic Statements	3.5.1
MM-4-2	Map TSP inputs	3.4.1
MM-4-3	Main TSP configuration screen	3.4.4
MM-4-4	For NTCIP/SCP operation, enter maximum extension and maximum reduction times	3.4.5
MM-5-2	Assign Special Function number to schedule to Inhibit TSP operation	3.4.6
MM-7-3-2	TSP/SCP Status Display	3.6
MM-8-1	Copy Utility to enable the TSP feature	3.1.1
MM-8-3	Enable TSP and SCP printout	3.8

After you enable the TSP special feature, there are three main tasks you will need to complete during TSP configuration. This chapter includes procedures with step-by-step instructions for each task:

- With **MM-4-2**, map the TSP inputs.
- With **MM-4-3**, select TSP or SCP and the default pattern number for Free Mode.
- With **MM-4-3**, identify the Transit Phases and settings.

3.4.1 Map TSP Signals

```

ENABLE PREEMPT FILTERING & TSP/SCP
FILTERED      SOLID      PULSING
INPUT  1  ...BYPASSED...  ...BYPASSED...
        2  ...BYPASSED...  ...BYPASSED...
        3  ..PREEMPT  3.  ..PREEMPT  7.
        4  ..PREEMPT  4.  ..PREEMPT  8.
        5  ..PREEMPT  5.  ..PREEMPT  9.
        6  ..PREEMPT  6.  ..PREEMPT 10.
        7  ...BYPASSED...  ...BYPASSED...
        8  ...BYPASSED...  ...BYPASSED...
        9  ...BYPASSED...  ...BYPASSED...
       10  ...BYPASSED...  ...BYPASSED...

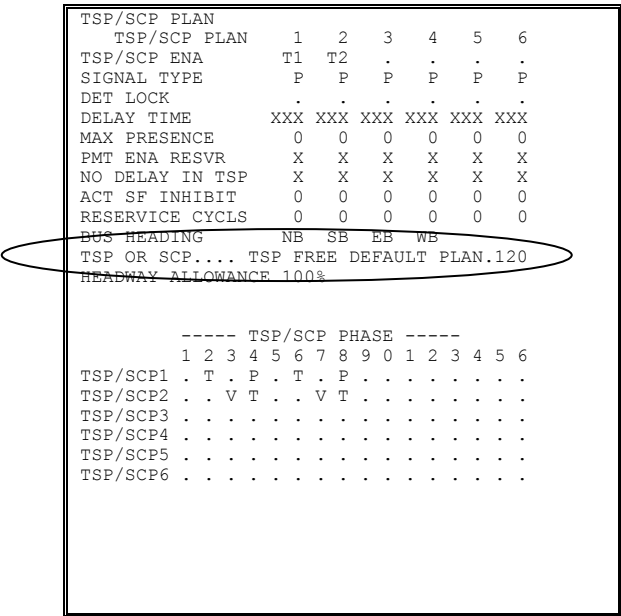
```

MM-4-2

ENABLE PREEMPT FILTERING & TSP/SCP

Prompt	Step	Procedure
FILTERED INPUT and SOLID	1	If you expect a solid TSP signal, move the cursor to the SOLID column next to the TSP input you want.
	2	Press [0] and toggle to “TSP/SCP IN 1” to select the TSP Check-In signal 1, etc.
	3	Press [0] and toggle to “TSP/SCP OUT 1” to select the TSP Check-Out signal 1, etc. IMPORTANT In MM-4-3 , for each TSP/SCP PLAN with SOLID signals, set SIGNAL TYPE = S. If SIGNAL TYPE = P, then these signals would NOT be connected.
FILTERED INPUT and PULSING	4	If you expect a pulsing TSP signal, move the cursor to the PULSING column next to the TSP input you want.
	5	Press [0] and toggle to “TSP/SCP 1” to select the TSP signal 1, etc. IMPORTANT In MM-4-3 , for each TSP/SCP PLAN with PULSING signals, set SIGNAL TYPE = P. If SIGNAL TYPE = S, then these signals would NOT be connected.

3.4.2 Select TSP/SCP Options



MM-4-3

TSP/SCP OPTIONS

Prompt	Step	Procedure
TSP OR SCP	1	Press [0] and toggle to TSP for ECPI, TSP operation OR SCP for NTCIP, SCP operation
FREE DEFAULT PTN	2	Enter a number (1-120) to select the number of the Free Default Pattern.

TSP Enable and Configuration

3.4.3 Create a New TSP Pattern for Coordinated Mode

To create TSP patterns, use the procedure below with **MM-3-2**, the **COORDINATOR PATTERN** submenu; you can use pattern numbers from 1 to 120 for TSP patterns during Coordinated Mode operation.

1. From the ASC/3 **MAIN MENU**, press **[3]**, **COORDINATOR**.
2. From the **COORDINATOR SUBMENU**, press **[2]**, **COORDINATOR PATTERN**.
3. In the **COORDINATOR PATTERN** submenu
 - A. At the **COORDINATOR PATTERN** prompt, enter a pattern number (1-120) that is not already in use.
 - B. Press **[ENTER]** to move to that pattern.

NOTE: Split Pattern is automatically defaulted 1 to 1 corresponding to the Coordinator pattern.

4. Enter information for the TSP pattern, as described in the following steps.

Pattern Number →

Cycle Length (0-999 sec) →

Phase Split for Coordinator Operation →

```
COORDINATOR PATTERN [ 1]
USE SPLIT PATTERN.. 1
TS2 (PAT-OFF).. 0-1
CYCLE..... 100s STD (COS).....111
OFFSET VAL..... 0s
ACTUATED COORD... NO TIMING PLAN.... 0
ACT WALK REST... NO SEQUENCE..... 0
PHASE RESRVCE... NO ACTION PLAN.... 0
SPLIT PREFERENCE PHASES
  PHASE[s] 1 2 3 4 5 6 7 8
  SPT[ 1] 0 0 0 0 0 0 0 0
  PREF 1... 0 0 0 0 0 0 0 0
  PREF 2... 0 0 0 0 0 0 0 0
  SPLT EXT. 0 0 0 0
  VEH PERM. 0 0 0 DISP
  RING DISP - 0 0 0 0 (RING 2-4)
SPLIT PREFERENCE PHASES
  PHASE[s] 9 10 11 12 13 14 15 16
  SPT[ 1] 0 0 0 0 0 0 0 0
  PREF 1... 0 0 0 0 0 0 0 0
  PREF 2... 0 0 0 0 0 0 0 0
  1 2
SPLIT DEMAND PTRN. xxx xxx XART PTRN. xxx

PHASE.. 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6
COORD.. . . . . . . . . . . . . . . .
VE RCALL . . . . . . . . . . . . . . .
PD RCALL . . . . . . . . . . . . . . .
MX RCALL . . . . . . . . . . . . . . .
OMIT.... . . . . . . . . . . . . . . .
SF OUT.. . . . . . . . . . (1-8)
```

MM-3-2 COORDINATOR PATTERN

5. Move the cursor down to the **CYCLE** prompt.
 - A. Press **[H]** and read the Help screen to determine the value of the cycle length. To avoid warning messages, the cycle length must be equal to or larger than the sum of the largest set of ring phases. Press **[H]** again to return to the data entry screen.
NOTE: Cycle split/relationship warning messages do not affect operation.
 - B. Set a value (0-999 seconds) for the cycle length.
6. Move the cursor to the Phase Split row.
 - A. Press **[H]** and read the Help Menu.
 - B. Enter the split times of the phases (the total time is the Cycle Length). In Free mode, the phase split is the Maximum green time of the phase.

NOTE: Data that you enter for a TSP Split Pattern for a Free Mode is always interpreted as being in seconds, never in percentages, even if you enter Coordinated Mode Split Pattern values in units of percent.

3.4.4 Programming TSP Plans and Parameters

NOTE: Refer to **Table 2-1** for descriptions of the *Configuration Parameters* in **MM-4-3**

```

TSP/SCP PLAN
  TSP/SCP PLAN      1    2    3    4    5    6
TSP/SCP ENA        YES YES  NO  NO  NO  NO
SIGNAL TYPE        P    P    P    P    P    P
DET LOCK           .    .    .    .    .    .
DELAY TIME         XXX XXX XXX XXX XXX XXX
MAX PRESENCE       0    0    0    0    0    0
PMT ENA RESVR      X    X    X    X    X    X
NO DELAY IN TSP    X    X    X    X    X    X
ACT SF INHIBIT     0    0    0    0    0    0
RESERVICE CYCLS   0    0    0    0    0    0
BUS HEADING        WB  EB  SB  NB
TSP OR SCP.... TSP FREE DEFAULT PTN.  1
HEADWAY ALLOWANCE  0%

      ----- TSP/SCP PHASE -----
      1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6
TSP/SCP1 . T . P . T . P . . . . .
TSP/SCP2 . . V T . . V T . . . . .
TSP/SCP3 . . . . . . . . . . . . .
TSP/SCP4 . . . . . . . . . . . . .
TSP/SCP5 . . . . . . . . . . . . .
TSP/SCP6 . . . . . . . . . . . . .

```

MM-4-3
TSP/SCP PLAN 1-6

Prompt	Step	Procedure
TSP/SCP ENA	1	Toggle to elect the desired operation for the TSP Plan: NO = TSP/SCP not enabled YES = Phasing as programmed in (MM-1-1-1) and (MM-1-1-2).
ACT SF INHIBIT	2	Enter a number (1-8) to specify which timebase Special Function will inhibit the TSP Plan. Use MM-5-2 to select and configure the Special Function.
DET LOCK	3	For a PULSING input, toggle to desired condition: X = Locks momentary pulsing input . = Does NOT lock momentary pulsing input
SIGNAL TYPE	4	Toggle to select the check-in detector signal: P = Enable detect 6.25 Hz Pulsing signals S = Enable detect single-pulse (solid) signals
DELAY TIME	5	For the TSP/SCP PLAN in this screen, enter the number of seconds (0-255) of delay between TSP Check-In and the start of the TSP movement.
RESERVICE CYCLS	6	Enter the number of cycles (0-255) that all TSP Inputs will be inhibited following a TSP movement or when any phase max or split has been altered.

Prompt	Step	Procedure
PMT ENA RESVR	7	Toggle to desired operation for the TSP Plan: X = TSP Reservice timer will start when preemption is terminated . = TSP Reservice timer will NOT start when preemption is terminated
NO DELAY IN TSP	8	Toggle to desired operation for the TSP Plan: X = Inhibit the Delay Time when a TSP phase is timing in green . = Call will not be serviced when Delay Time is timing
MAX PRESENCE	9	“Presence” refers to “time.” For the TSP/SCP PLAN in this screen, enter the maximum time in seconds (0-999) that a TSP call can be held OR that a TSP movement can be active before timeout.
TSP/SCP PHASES SECTION	10	For each TSP/SCP plan, each phase can be assigned to one of the options below: . = No programming T = TSP/SCP Phase. referred to as Priority Strategy Service (TSP) Phases V = Phase Omit. This is a per TSP plan parameter. The Coordinator shall ignore any omit applied to the coordinated phase(s). This phase omit is logically ORed with any other phase omits that may be active. P = Ped Omit. This is a per TSP plan parameter. This ped omit is logically ORed with any other ped omits that may be active. These ped omits do not apply to non-actuated phases.

TSP Enable and Configuration

3.4.5 Programming TSP/SCP Split Patterns

MM3-2

```
COORDINATOR PATTERN [ 1]
USE SPLIT PATTERN...
TS2 (PAT-OFF)... 0-1
CYCLE..... 100s STD (COS).....111
OFFSET VAL..... 0s
ACTUATED COORD... NO TIMING PLAN.... 0
ACT WALK REST.... NO SEQUENCE..... 0
PHASE RESRVC.... NO ACTION PLAN.... 0
SPLIT PREFERENCE PHASES
  PHASE[s] 1 2 3 4 5 6 7 8
SPT[ 1] 0 0 0 0 0 0 0 0
PREF 1... 0 0 0 0 0 0 0 0
PREF 2... 0 0 0 0 0 0 0 0
SPLT EXT. 0 0 0 0
VEH PERM. 0 0 0 DISP
RING DISP - 0 0 0 (RING 2-4)
SPLIT PREFERENCE PHASES
  PHASE[s] 9 10 11 12 13 14 15 16
SPT[ 1] 0 0 0 0 0 0 0 0
PREF 1... 0 0 0 0 0 0 0 0
PREF 2... 0 0 0 0 0 0 0 0
SPLIT DEMAND PTRN. 1 2 XXX XART PTRN. XXX
```

MM3-3

```
SPLIT PATTERN [ 1]
PHASE[s] 1 2 3 4 5 6 7 8
SPIT 0 0 0 0 0 0 0 0
PHASE[s] 9 10 11 12 13 14 15 16
SPIT 0 0 0 0 0 0 0 0
PHASE.. 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6
COORD... . . . . .
VE RCALL . . . . .
PD RCALL . . . . .
MX RCALL . . . . .
OMIT.... . . . . .
```

MM4-4

```
TSP/SCP SPLIT PATTERN [ 1]
IN EFFECT TMG PLAN [#] 0 SPL DM [#] 0
PHASE 1 2 3 4 5 6 7 8
MAX RDTN.255 255 255 255 255 255 255 255
MIN GRN.. 5 5 5 5 5 5 5 5
MAX EXTN.255 255 255 255 255 255 255 255
PHASE 9 10 11 12 13 14 15 16
MAX RDTN.255 255 255 255 255 255 255 255
MIN GRN.. 5 5 5 5 5 5 5 5
MAX EXTN.255 255 255 255 255 255 255 255
```

Associate Coordinator Pattern with any split Pattern

1 to 1 Association between MM-3-3 and MM-4-4

When TSP is active, TSP operation adjusts the current split (**MM-3-3**) through MAX RDTN and MAX EXTN parameters, programmed in **MM-4-4**. The current split (**MM-3-3**) in the current Coordinator Pattern (**MM-3-2**) can be invoked by Time of Day or other sources.

For example, if the current Phase split is 20 seconds (**MM-3-3**), and the MAX RDTN in the associated phase for the current phase split (**MM-4-4**) is 5 seconds, then the TSP minimum split time is $20 - 5 = 15$ seconds.

MM-4-4 programs the TSP or SCP split extension of the coordination **SPLIT PATTERN** in effect (specified in **MM-3-2**).

1. Read the notes below for your application.
2. Use the instructions on the subsequent page to enter data.

TSP Operation

1. In **MM-4-3**, be sure that **TSP OR SCP = TSP**.
2. Go to Phase Max Reduction, **MAX RDTN**, and enter the maximum number of seconds that the split of a phase can be reduced during a priority strategy.
 - a. The TSP minimum splits for phases associated with TSP are computed by subtracting each **MAX RDTN** value from its coordination split.
 - b. The TSP split maximums are computed from the cycle length for each ring.
 - i. The length of the longest timing ring is subtracted from the specified cycle length.

- ii. The difference in (i) is added to the minimum TSP splits of the TSP phases.
3. For Coordination Mode, **MIN GRN** (TSP MIN) shows the result of the calculation of the Coord Split (**MM 3-2**) minus the **MAX RDTN** (**MM 4-4**) = TSP Min Green + Clearance Time.

SCP Operation

1. In **MM-4-3**, be sure that **TSP OR SCP = SCP**.
2. Go to Phase Max Extension, **MAX EXTN**, and enter the maximum number of seconds that the split of a phase can be extended during a priority strategy.
 - a. The effective value may be modified by the coordination based on the time gained from the reductions of the non-priority phases.
 - b. This is a per-phase and per-split pattern parameter that specifies the maximum number of seconds that the split of a phase can be extended during SCP.
3. Go to Phase Max Reduction, **MAX RDTN**, and enter the maximum number of seconds that the split of a phase can be reduced during a priority strategy.
 - a. The effective value may be modified by the coordination based on the operating phase minimum service requirements.
 - b. NOTES:
 - A phase minimum time is determined if a phase can be terminated or forced off by the coordinator.
 - This is a per-phase and per-split pattern parameter that specifies the split of a phase can be reduced during SCP.
 - This effective value may be reduced based on the operating phase minimum service requirements.

TSP Enable and Configuration

MM-4-4

```
TSP/SCP SPLIT PATTERN [ 1]
IN EFFECT TMG PLAN [#] 0 SPL DM [#] 0
  PHASE  1  2  3  4  5  6  7  8
MAX RDTN.255 255 255 255 255 255 255 255
MIN GRN..  5   5   5   5   5   5   5   5
MAX EXTN.255 255 255 255 255 255 255 255

  PHASE  9 10 11 12 13 14 15 16
MAX RDTN.255 255 255 255 255 255 255 255
MIN GRN..  5   5   5   5   5   5   5   5
MAX EXTN.255 255 255 255 255 255 255 255
```

MM-3-3

```
SPLIT PATTERN [ 1]
  PHASE[s]  1  2  3  4  5  6  7  8
SPIT       0  0  0  0  0  0  0  0

  PHASE[s]  9 10 11 12 13 14 15 16
SPIT       0  0  0  0  0  0  0  0

PHASE..  1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6
COORD...  . . . . . . . . . . . . . .
VE RCALL . . . . . . . . . . . . . .
PD RCALL . . . . . . . . . . . . . .
MX RCALL . . . . . . . . . . . . . .
OMIT....  . . . . . . . . . . . . . .
```

These 2 screens are 1 to 1 associated (e.g Pattern 2 on MM-4-4 is associated with Pattern 2 on MM-3-3)

MM-4-4 TSP/SCP SPLIT PATTERN

This screen is used for both TSP and SCP operation. Refer to notes on the previous page.

SCP (Signal Control and Prioritization) in NTCIP is the equivalent of TSP (Transit Signal Priority) for ECPI.

Prompt	Step	Procedure
TSP/SCP SPLIT PATTERN	1	Enter the number (1-120) of the Split Pattern: - For Coordinated Mode, enter the SPLIT PATTERN number from MM3-2. - For Free Mode, enter the FREE DEFAULT PTN number from MM-4-3.
TSP/SCP MAX RDTN	2	For each phase (1-16), enter the maximum number of seconds (0-255) the split of a phase can be reduced during TSP.
MIN GRN	3	Read the result of the calculation of the coord split (MM 3-3) minus the TSP/SCP MAX RDTN (MM 4-4) = TSP Min Green + Clearance Time
SCP MAX EXTN	4	For each phase (1-16), enter the maximum number of seconds (0-255) the split of a phase can be extended during SCP.

3.4.6 Inhibit TSP through Special Function.

Special Function in **MM-5-2 – Action Plan**, provides the means to inhibit TSP

Special Function
for TSP Inhibit

```

ACTION PLAN...[   ]
PATTERN.....      SYS OVERRIDE....YES
TIMING PLAN.....  SEQUENCE.....
VEH DETECTOR PLAN. DET LOG.....NONE
FLASH..... EN RED REST.....YES
VEH DET DIAG PLN.. PED DET DIAG PLN..
DIMMING ENABLE..

PH-FUNC  1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6
PED RCL.  . . . . .
WALK 2..  . . . . .
VEX 2...  . . . . .
VEH RCL.  . . . . .
MAX RCL.  . . . . .
MAX 2...  X X X X X X X X . . . . .
MAX 3...  . . . . .
CS INH..  . . . . .
OMIT.....  . . . . .
SPC FCT.  . . . . . (1-8)
AUX FCT.  . . . . (1-3)

LP 1-15   . . . . .
LP 16-30  . . . . .
LP 31-45  . . . . .
LP 46-60  . . . . .
LP 61-75  . . . . .
LP 76-90  . . . . .
LP91-100  . . . . .

```

**MM-5-2
ACTION PLAN**

Prompt	Step	Procedure
SPC FCT	1	Move the cursor down to the SPC FCT prompt, as shown above. SPC FCT = Special Function
	2	To Enable/Disable a Special Function, position the cursor under its number (1-8).
	3	The number of the Special Function corresponds to the number entered in ACT SF INHIBIT in MM-4-3 . Toggle to Enable/Disable the Time Base Special Function (1-8): X = Enable the Action Plan configured on this screen to output this Time Base Special Function . = Disable the Action Plan configured on this screen to output this Time Base Special Function Special Function 3 is enabled in the example screen above. The Time Base Special Function will inhibit TSP operation during the times specified by the Action Plan.

3.5 Logic Processor Statements

You can use the Logic Processor to command the controller inputs and outputs based on a set of logical statements.

In the *ASC/3 Programming Manual* (100-0903-001), refer to **Paragraph 6.12.1, Logic Processor Statement Control**; **Paragraph 6.12.3, General Programming Notes for Logic Processor Statements**; and **Appendix L/Chapter 26, Logic Processor Operation**.



When you program the Logic Processor, the controller must be on the bench and not operating an intersection.

3.5.1 Procedure to Program Logic Processor Statements

Use **MM-1-8-2, LOGIC STATEMENTS**, to program logic statements to control the 6 TSP inputs and 6 TSP outputs.

```

LP#  1  COPY FROM:  1  ACTIVE: N
IF    DETECTOR                1 IS ON
THEN SET TSP CHKIN DET      1   ON
ELSE SET TSP CHKIN DET      1   OFF
  
```

MM-1-8-2

LOGIC PROCESSOR STATEMENTS (LPS)

Prompt	Step	Procedure
LP#	1	Enter the number for this logic statement, 0 - 100.
ACTIVE	2	You can activate MM-1-8-2 LPS by different sources. The source of the activation is shown on the ACTIVE field: • A = Enabled by Action PLAN • M = Enabled by MM-1-8-1 (Manual) • R = Enabled by Remote command N = Not Enabled

Prompt	Step	Procedure
IF	3	<i>Conditional Statement:</i> 1. Press [ENTER] to go to an expanded Select List. 2. Use the arrows to go to the desired element. 3. Press [ENTER] again. Refer to the list of Testable Elements in <i>Appendix L</i> of the <i>ASC/3 Programming Manual</i>.
THEN	4	<i>Executable Statement:</i> 1. Press [ENTER] to go to an expanded Select List. 2. Use the arrows to go to the desired element. 3. Press [ENTER] again. Refer to the list of Executable Elements in <i>Appendix L</i> of the <i>ASC/3 Programming Manual</i>.
ELSE	5	<i>Executable Statement:</i> 1. Press [ENTER] to go to an expanded Select List. 2. Use the arrows to go to the desired element. 3. Press [ENTER] again. Refer to the list of Executable Elements in <i>Appendix L</i> of the <i>ASC/3 Programming Manual</i>.

3.6 Read MM-7-3-2, TSP/SCP STATUS DISPLAY

```

Line
1  TSP/SCP[FREE-TBC      ]02/12/08|07:09:40
2  PHASE 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6
3  PH STAT R G R R G R R R - - - - -
4  VEH OVL - - - - - - - - - - -
5  PED OVL W W W W W W W W W W W W W
6  VEH CALL . I I . . . . .
7  PED CALL . . . . .
8  ACT TSP/SCP[0]| STATUS 1H 2H 3H 4H 5H 6H
9  PLAN SPLT:12|TP:4|SEQ:1|ACT:100|DP:4
10 LC: 6s/ 0|SC: 0s[ FREE]TLM ADD: 0
11 R1/PH 4|R2/PH 8|R3/PH .|R4/PH .
12 MGR1 0.0|YEL 0.0|RED REST |RED REST
13 PDCL 12.0|FORCE OFF| |
14 FUNCTION 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6
15 PMT/TSP 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
16 LP FLAG . A . . . A . . . . .

```

When in a black hyperlink field:

To Go to the Hyperlinked Display

For an ASC/3:

1. Press [SPEC FUNC].
2. Press [NEXT DATA].

For an ATC 2070:

1. Press [*].
2. Press [D].

To Go Back to the Original Display

- For an ASC/3, press [SUB MENU].
- For an ATC 2070, press [ESC].

TSP/SCP Field Testing

To test operation and place a manual TSP/SCP call:

1. Go to **MM-7-3-2**.
2. Move the cursor to the applicable TSP input field on the PMT/TSP row.
3. Press [ENTER] and hold the key down. This will apply a continuous pulsing signal to the designated TSP input, simulating a TSP call.

Active Plan or State — TSP/SCP Status, MM-7-3-2, Line 1

Parameter	Definition	Parameter	Definition
[FAULT-MMU/ERR]	MMU response errors. Check MM-7-7-2.	[FAULT-CTR/ERR]	Critical error. Check MM-7-7-1
[FAULT-CTR/BIU]	TF BIU 1-4 error. Check MM-7-7-2.	[FAULT-PMT/PUP]	Preemptor sensed during Start-Up. Deactivate preempt to exit flash.
[FAULT-PMT/LCK]	Preempt, Interlock fail for 3 possible causes: • Preemptor and Interlock inputs both are ACTIVE simultaneously, even after Input Delay Time. • Interlock input is TRUE before 1 second was up. • Both Preemptor and Interlock inputs are INACTIVE simultaneously.	[FAULT-PMT/CRC]	Preempt DB CRC not matched (MM-1-7-1). Need to change DB or rewire hardware CRC inputs.
[FLSH-LOCAL FL]	Local Flash detected [FLSH- ST MMU] MMU detected error. Stop time applied with Flash. Check MMU for error information.	[FLSH- ST MMU]	MMU detected error. Stop Time applied with Flash. Check MMU for error information.
[FLSH- CTR CVM]	Controller caused Flash. Verify that the database matches the controller type. For example, the 2070 A3000.db requires the 2070-2A module.	[FLSH- CTR TBC]	Flash by Time Base Control. Check active Action Plan.
[FLSH- CTR SYS]	Flash by Coord Pattern. Check active Coordination Pattern.	[FLSH MMU INPT]	MMU detects AUTO/FLASH input active.

TSP Enable and Configuration

Parameter	Definition	Parameter	Definition
[PMTxx DLY yyy]	Preemptor xx is timing down its Delay Timer yyy. When it reaches ZERO, the preempt will start.	[PMTxx INH yyy]	Preemptor xx is in its INHIBIT interval yyy. VEH/PED calls received during this interval will not be serviced until after the preemptor exits.
[PMTxx TC yyy]	Track Clear Timing: Preemptor xx is timing its Track Clear Grn, Yel or Red	[PMTxx TCMXyyy]	Preemptor xx Track Clear Max timer
[PMTxx DWL yyy]	Preemptor xx is timing down its Minimum Dwell time. Preempt may not exit until this timer reaches ZERO.	[PMTxx MAX yyy]	Preemptor xx Max timer
[PMTxx FLSHyyy]	Preemptor xx is timing down its Minimum Dwell Flash time. Preempt cannot exit until this timer reaches ZERO.	[PMTxx EXT2CRD]	Preemptor xx is timing down its Minimum Dwell Time. Preempt cannot exit until this timer reaches ZERO.
[PMTxx EXT2PHS]	Pmt x is attempting to exit to phases as directed by the coordinator	[PMTxx EXTNONE]	Preemptor xx is exiting with no exit phases selected.
[MNL CT ENA IA]	Manual Control Enable, Interval Advance	[STOP TIME]	Stop Time input is applied.
[xxx UPLOAD]	Port xxx is uploading.	[xxx DOWNLOAD]	Port xxx is downloading.
[CRD SYS Pyyy]	In Coordination. Pattern yyy is active and is commanded by system software.	[CRD TBC Pyyy]	In Coordination. Pattern yyy is active and is commanded by Time Base Control.
[CRD MAN Pyyy]	In coordination. Manual pattern yyy is active.	[CRD FREE ALM]	In Coordination. Incorrect parameter found in the commanding pattern.
[CRD yyFO ERR]	In Coordination. Phase yy failed to be Forced-Off	[CRD LZ ERR]	In Coordination. Coordinated phase is not active at local zero.
[FREE MAN]	In free. Manual free pattern is in effect.	[FREE INPUT]	In Free. Commanded by HW Input
[FREE SYS]	In Free. Commanded by System source.	[FREE TBC]	In Free. Commanded by Time Base Control.
[CRD TSPxSyyy]	In Coordination. TSP call x and split pattern yyy are active.	[FREE TSPxSyyy]	In Free. TSP call x and split pattern yyy are active.
[PMTxx GATEyyy]	Gate Down timer: Preemptor xx is timing down its Extend Track Green Timer yyy. NOTE: The timer will not start to time down until after the GATE DOWN input has occurred.		

Other Entries on TSP/SCP Status, Line 1

- Current Controller *Date* in MM/DD/YY format
- Current Controller *Time* in HH:MM:SS format

Phase Status, in Order of Precedence — TSP/SCP Status, MM-7-3-2, Line 3

PH STAT	Definition
—	Phase is NOT enabled
N	Phase in Next
R	Phase is timing Red
Y	Phase is timing Yellow

TSP Enable and Configuration

PH STAT	Definition
y (lower case y) special	Protected Permitted Left Turn (PPLT) yellow
D	Phase is timing delay green
.	Phase is Red and not timing
G	Phase is Green
X	Simultaneous Red/Yellow (orange)
F# (special)	Canadian FF (future native feature) F1, F2, F3

Vehicle Overlaps — TSP/SCP Status, MM-7-3-2, Line 4

VEH OVL	Definition
-	Overlap is not programmed (omitted)
A	Overlap is timing Advance green
N	Overlap is next
Y	Overlap is timing Yellow
y (special)	Protected Permitted Left Turn (PPLT)
G	Overlap is green
F#	Flashing Green Overlap
L	Overlap is Timing Lagging green
R	Overlap is Red

Pedestrian Overlap Status (order of precedence) — TSP/SCP Status, MM-7-3-2, Line 5

PED OVL	Definition
-	PED is not enabled
N	PED is next
W	PED is timing/in WALK
C	PED is timing PED CLEAR
O	PED is Omitted
D	PED is in DON'T WALK

Vehicle Call Information — TSP/SCP Status, MM-7-3-2, Line 6

VEH CALL	Definition
O	Phase Omitted
E	Red extension call
C	Active vehicle detector call
B	Bike call
I	Call placed by internal application (Power Up, Preemptor, Keyboard input)
N	Phase is non-actuated (CNA1, CNA2, Pre-timed)
F	Call placed by detector diagnostic (Failed Detector)
M	Max recall programmed for phase
R	Vehicle recall programmed for phase
X	Call placed by Max out
L	Locked vehicle detector call
S	Call placed by Soft recall or do not rest here
.	No phase demand

Vehicle Calls. To make a Vehicle Call, go to MM-7-3-2, Line 4, VEH CALL and obey the applicable procedure, below.

To Make a Vehicle Call on:	Procedure
All Phases	1. Move the cursor to the second L of VEH CALL 2. Press [ENTER] .
One Phase (range 1 thru 16)	1. Move the cursor to the applicable phase, 1 thru 16. 2. Press [ENTER] .
One Phase (range 1 thru 10)	1. Move the cursor to anywhere in the VEH CALL line. 2. Press the applicable number key [1] thru [0] .

NOTE: You can use the same procedure, above, to make a Vehicle Call from any status display that has the VEH CALL parameter.

Pedestrian Call Information — TSP/SCP Status, MM-7-3-2, Line 7

PED CALL	Definition
O	Ped Omitted
C	Active Pedestrian Detector Call
2	Active Pedestrian 2 Detector Call
I	Call placed by internal application (Power Up, Keyboard input, Logic)
N	Pedestrian Phase is non-actuated (CNA1, CNA2, Pre-timed)
F	Call placed by detector diagnostic (Failed Detector)
R	Pedestrian Recall
X	Locked pedestrian call on the phase
.	No Pedestrian demand

Pedestrian Calls. To make a Pedestrian Call, go to MM-7-3-2, Line 5, PED CALL and obey the applicable procedure, below.

To Make a Pedestrian Call on:	Procedure
All Phases	1. Move the cursor to the second L of PED CALL 2. Press [ENTER] .
One Phase (range 1 thru 16)	1. Move the cursor to the applicable phase, 1 thru 16. 2. Press [ENTER] .
One Phase (range 1 thru 10)	1. Move the cursor to anywhere in the PED CALL line. 2. Press the applicable number key [1] thru [0] .

NOTE: You can use the same procedure, above, to make a Pedestrian Call from any status display that has the PED CALL parameter.

Active TSP/SCP Plan — TSP/SCP Status, MM-7-3-2, Line 8 (left side)

Parameter	Definition	Range
ACT TSP/SCP	Number of the active TSP/SCP plan	1 thru 6

TSP/SCP Plan Status — TSP/SCP Status, MM-7-3-2, Line 8 (right side)

Parameter	Definition	Range
STATUS	TSP Status for the TSP plans	X = D, P, I, O, H, or E
1X	D = TSP/SCP Disabled	
2X	P = No valid TSP/SCP Pattern	
3X	I = TSP Phase inhibited	
4X	O = Omits on TSP/SCP phases	
5X	H = No TSP/SCP phases	
6X	E = Enabled and Idle	

TSP Enable and Configuration

Current Controller Configuration — TSP/SCP Status, MM-7-3-2, Line 9

Hyperlink Fields

Parameter	Definition	Range
PLAN SPLT:XXX	Plan Split Pattern Number	XXX = 001 thru 120
TP:XX	Timing Plan Number	XX = 01 thru 04
SEQ:XX	Sequence Page Number	XX = 01 thru 16
ACT:XXX	Action Plan Number	XXX = 001 thru 100
DP:X	Detector Plan Number	X = 1 thru 4

Coordination-Specific Timing Information, TSP/SCP Status, MM-7-3-2, Line 10

These are in four separate areas, as show below: Local Cycle Counter,
System Cycle Counter, Current Coordination Status, Highest Priority Telemetry Address

System Cycle Counter, Current Coordination Status, Highest Priority Telemetry Address				
Parameter	Definition	Ranges		
LC:XXX+/YYY	Local Cycle Counter	XXX = counter	+ = s for seconds or % for percent	YYY = actual cycle length programmed
SC:XXX+	System Cycle Counter (Master Clock)	XXX = counter	+ = s for seconds or % for percent	
Current Coordination Status				
FREE	Free	N/A		
SYNC	Sync Pulse			
PICKUP	In pickup mode			
COORD	Coordinated (in step)			
DWELL	Dwelling / Transition			
ADDONLY	Add Only Transition			
DEDUCT	Subtracting in Smooth Transition			
ADDING	Adding in Smooth Transition			
Highest Priority Telemetry Address				
TLM ADD XXXXX	Highest Priority Telemetry Address	XXXXXX = 0 thru 65535		

Ring and Active Phase for the Ring — TSP/SCP Status, MM-7-3-2, Line 11

If a Ring is NOT active, the phase is a “.”

Parameter	Definition	Ranges	
RX/PH YY	Ring Number/Active Phase	X = 1 thru 4	YY = 1 thru 16

For example, R1/PH 4 | R2/PH 8 | R3/PH . | R4/PH .

shows that Ring 1 has active phase 4| Ring 2 has active phase 8| Ring 3 is not active| Ring 4 is not active

Active Phase Timing and Intervals — TSP/SCP Status, MM-7-3-2, Line 12

NOTE: For a timer, the timer value is shown to its right.

Parameter	Definition
EXT1	Green vehicle extension
EXT2	Green vehicle extension 2

Parameter	Definition
GAPr	Gap reduction is in effect
GRNp	Preemptor minimum green override
GRN HOLD	Phase in in HOLD interval
GRN REST	Phase rests in absence of calls
GRN XFER	Phase awaits timeout of another phase
INACTIVE	Ring is inactive
MGRN1	Minimum Green
MGRN2	Bike minimum green
MGRa	Additional Initial Green
MGRNd	Delay green for early walk
MGRNc	Conditional Service minimum green
MGRNg	Guaranteed minimum green
PED ONLY	Phase is timing pedestrian timing only
RED–	Vehicle red clearance
RED REST	Red rest
REDg	Guaranteed red clearance
REDp	Preemptor red override
RED EXTEND	Phase is timing red extension
RED XFER	Phase is in red transfer
RRVT	Phase is next but its red revert is timing
YEL–	Vehicle yellow change
YELg	Guaranteed yellow change
YELp	Preemptor yellow override

Phase Termination Timing, TSP/SCP Status, MM-7-3-2, Line 13

When a phase is terminated (during yellow change and red clearance),
the reason for the termination is shown here.

Parameter	Definition
FORCE OFF	Terminated by command or force-off signal
GAP OUT	Gap between vehicles exceeded gap-out limit in effect in volume density operation
GUAR PASS	Full vehicle extension timed after time out of reduced gap
MAN ADV	Terminated by manual advance
MAX1	Max 1 is timing
MAX2	Max 2 is timing
MAX3	Max 3 is timing
MAXD	Max Dynamic is timing
MAX OUT	Green ext time reached its maximum limit
MAX GAP	Minimum gap was in effect when gap-out occurred
PC–1	Ped Clear 1 is timing
PC–2	Ped Clear 2 is timing
PC–G	Guarantee Ped Clear is timing
PC–M	Ped Clear Max is timing
PC–P	Preemptor is timing Ped Clear
PREEMPT	Terminated by preemption
REDm	Vehicle red clearance maximum

TSP Enable and Configuration

Parameter	Definition
SPLT	Coord Split is timing
TSP	TSP Split is timing
WK-1	Walk 1 is timing
WK-G	Guarantee Walk is timing
WALK HOLD	The phase is in Walk Hold
WALK REST	The Phase is in Green Reset
WK-2	Walk 2 is timing
WLFT	Pre Time Walk (FT) is timing
WL-P	Preemptor is timing walk
WK-M	Walk max is timing

Preemptor (PMT) & Transit Signal Priority (TSP) Input Status

TSP/SCP Status, MM-7-3-2, Line 15

10 possible Preempts (1 thru 10) are on the left, and 6 possible TSPs (1 thru 6) are on the right

PMT/TSP	Definition
PMT Status (left 1 thru 10)	
-	Not Programmed
R	Re-service period
I	Preemptor is Inhibited
D	Delay timer is running for that preempt
A	Preempt run is active
C	Call, but that preempt is not yet active
.	Preempt is programmed/enabled, but there is no activity on it
TSP Input Status (right 1 thru 6)	
-	Disabled
X	TSP is not programmed correctly (omits on TSP phases, no TSP pattern, no TSP phases)
R	Re-service period
I	Inhibited
A	Call is being served
N	NTCIP Commanded
C	Call from inputs
.	Enabled, but there is no activity on it
G	A TSP phase is timing in early or Extended Green

Logic Processor Flag Status, TSP/SCP Status, MM-7-3-2, Line 16

16 possible Logic Processor Flags, 1 thru 16

LP FLAG	Definition
A	Active
.	Inactive
-	Not used

3.7 Metro Rapid DTGP Action Response Bits

The bits in the table below will be included in the response message in field #10 of the Metro Rapid Protocol.

DTGP = Decision to Grant Priority

Bit #	Name	Description
1	Hold green phase	The TSP movement is in effect and a bus phase has been extended.
2	Early green return	The TSP movement is in effect and a bus phase has been in early green return.
3	Headway Override	Priority request is denied based on headway adherence Calculation (for Metro Protocol only)—that is, actual headway is less than (schedule headway + headway allowance) where headway allowance = schedule headway x allowance percentage (MM-4-3).
4	Demand Override	By Time-of-Day ONLY (priority request is inhibited by SPC FCT (1-8) defined in action plan MM-5-2 and ACT SF INHIBIT in MM-4-3).
5	Data Override	Priority is denied due to non-matching city and address codes.
6	Reservice Override	Priority request is denied because the reservice cycle (MM-4-3) is in effect.
7	Priority Override	Priority request is denied because a priority request is being serviced.
3 & 8	Offset Transition Override	Priority request is denied because the controller is in offset transition.
4 & 8	Flash Override	Priority request is denied because the controller is in flash.
5 & 8	Configuration Override	Priority request is denied because there is an invalid configuration.
7 & 8	Preempt Override	Priority request is denied because a preemption request is being serviced.
6 & 8	Other Overrides	Priority request is denied because of reasons not listed above.

3.8 Print Out TSP/SCP Information from PRINT MM-8-3

```

PRINT

PRINT TO PORT... 2

                        SELECT ALL..... .

CONFIGURATION... . TIME BASE..... .

CONTROLLER..... . DETECTOR..... .

COORDINATOR..... . LOGIC PROCESSOR. .

PREEMPTOR..... . TSP AND SCP..... .

TOGGLE TO SELECT AND THEN PRESS ENTER
TO BEGIN PRINTING

```

MM-8-3 PRINT

Prompt	Step	Procedure
PRINT TO PORT	1	Enter the number of the port to which to send the printed information.
TSP AND SCP	2	Press [0] to toggle to: X = Enable TSP or SCP information to print out OR . = Disable TSP or SCP information from printing out
	3	Press [ENTER] to start printing.
	4	The printed information goes to the selected port.

3.9 Enable TSP Event Logging

```
EVENT LOGGING
RFEs (MMU/TF).. YES 3 RFes >24H..... YES
MMU FL FAULTS.. YES LOCAL FLASH..... YES
RFEs (DET/TEST) YES DETECTOR ERR..... YES
COORD ERRORS... YES CTR DOWNLOAD.... YES
PREEMPT..... YES TSP..... YES
POWER ON/OFF... YES LOW BATTERY..... YES
ACCESS..... YES DATA CHANGE..... YES
ALARM 1..... YES ALARM 2..... YES
ALARM 3..... YES ALARM 4..... YES
ALARM 5..... YES ALARM 6..... YES
ALARM 7..... YES ALARM 8..... YES
ALARM 9..... YES ALARM 10..... YES
ALARM 11..... YES ALARM 12..... YES
ALARM 13..... YES ALARM 14..... YES
ALARM 15..... YES ALARM 16..... YES
```

MM-1-6-1
EVENT LOGGING

Prompt	Step	Procedure
TSP	1	Press [0] to toggle to: YES = Enable logging of 4 possible TSP events: TSP Call Received TSP Inhibited Call Received TSP Cycle Activated TSP Cycle Terminated OR NO = Disable logging of TSP events. NOTE: MM-8-6-1-1 displays the TSP events.

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